

PROJECT REPORT OF CO-OP Project at Industry (Module-2)

ON

Digital Memory Decay

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled “**Digital Memory Decay**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of Dr Gurpreet Singh. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura

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Date: 18th May 2026

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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INTRODUCTION

The rapid growth of digital storage systems has resulted in massive accumulation of personal digital archives such as documents, photographs, emails, videos, chats, and cloud backups. Over time, these archives experience degradation due to corruption, storage failure, data redundancy, hardware limitations, and accessibility issues.

The project “Digital Memory Decay: Modeling Information Loss in Personal Data Archives” is designed to simulate and analyze long-term information degradation using computational analytics and predictive modeling techniques.

The system evaluates archive preservation conditions, corruption exposure, access frequency, and storage reliability to estimate preservation scores and risk levels associated with digital archives.

The project also provides interactive dashboard visualizations, comparative analysis reports, recommendation systems, and heatmap analytics for improved archive preservation assessment.

OBJECTIVES

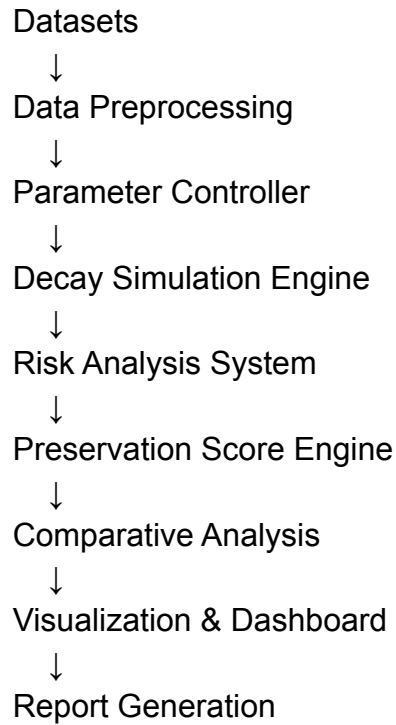
- To model digital archive degradation over time.
- To analyze preservation behavior using computational simulation.
- To evaluate corruption risk and storage reliability.
- To generate preservation scores for digital archives.
- To develop a web-based dashboard for archive analytics.
- To generate visual reports and comparative analysis.
- To provide recommendations for archive preservation strategies.

PROBLEM STATEMENT

Personal digital archives are highly vulnerable to corruption, storage degradation, accidental deletion, hardware failure, and reduced accessibility over time. Existing storage systems often lack predictive analysis mechanisms to estimate archive retention and preservation quality.

This project aims to address these challenges by developing a predictive analytical system capable of simulating information decay and evaluating long-term archive stability.

SYSTEM ARCHITECTURE



TECHNOLOGIES USED

- Python
- Flask Framework
- HTML
- CSS
- Pandas
- NumPy
- Matplotlib
- Seaborn

MODULES OF THE PROJECT

1. Data Preprocessing Module

Handles dataset loading, formatting, and preprocessing.

2. Parameter Controller

Controls archive decay parameters based on corruption risk, storage condition, and backup status.

3. Decay Simulation Engine

Simulates archive degradation over time using predictive computational modeling.

4. Risk Analysis System

Determines archive risk levels including Low, Moderate, and Critical.

5. Preservation Score Engine

Calculates preservation scores and evaluates archive stability.

6. Comparative Analysis Module

Compares multiple datasets and generates analytical visualizations.

7. Recommendation Engine

Provides preservation recommendations based on archive condition.

8. Dashboard System

Displays interactive analytics and archive management visualizations.

DATASET DESCRIPTION

The project uses five simulated datasets:

- sample_dataset_1.csv
- sample_dataset_2.csv
- sample_dataset_3.csv
- sample_dataset_4.csv
- sample_dataset_5.csv

Each dataset contains information regarding:

- Archive ID
- File Type
- Access Frequency
- Storage Condition
- Backup Status
- Initial Files
- Corruption Risk
- Years Stored

METHODOLOGY

The system first loads digital archive datasets and preprocesses them for computational analysis. The parameter controller calculates decay rates using storage condition, corruption exposure, access frequency, and backup reliability.

The decay engine then simulates long-term archive degradation using predictive mathematical modeling. Based on simulated retention results, preservation scores and risk levels are generated.

Finally, analytical visualizations, dashboard interfaces, heatmaps, and comparative reports are generated for archive evaluation.

RESULTS AND ANALYSIS

The project successfully generated:

- Archive preservation analytics
- Comparative dataset analysis
- Heatmap visualization
- Dashboard monitoring system
- Recommendation reports
- Risk assessment reports

- Preservation score evaluation

The system demonstrated effective simulation of long-term information degradation and archive retention forecasting.

CONCLUSION

The project successfully modeled digital information decay in personal data archives using computational analytics and predictive simulation techniques.

The system effectively analyzed archive preservation behavior, generated preservation scores, evaluated risk levels, and provided visual insights through interactive dashboards and comparative reports.

The developed framework can assist in understanding long-term digital preservation challenges and archive management strategies.

FUTURE SCOPE

Future improvements may include:

- Integration with cloud storage APIs
- Machine learning-based prediction models
- Real-time archive monitoring
- Advanced cybersecurity analysis
- User authentication systems
- Database integration
- Automated backup management systems